

Experiment - 13

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Course Code: MLA0305

Slot: B

Aim

To implement and compare **PPO, TRPO, and DDPG** reinforcement learning algorithms and evaluate their performance using the Pendulum environment.

Algorithm / Procedure

1. Initialize the Pendulum environment.
2. Create PPO, TRPO, and DDPG agents.
3. Train each agent for a fixed number of time steps.
4. Run each trained agent in the environment.
5. Record the reward obtained by each agent.
6. Calculate the average reward for each algorithm.
7. Compare the algorithms using a performance graph.
8. Identify the algorithm with better performance.

Code

```
!pip -q install gymnasium stable-baselines3 sb3-contrib
```

```
import gymnasium as gym
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from stable_baselines3 import PPO, DDPG
```

```
from sb3_contrib import TRPO
```

```
# Create environment
```

```
env = gym.make("Pendulum-v1")
```

```
# Create agents
```

```
ppo = PPO("MlpPolicy", env, verbose=0)
```

```
trpo = TRPO("MlpPolicy", env, verbose=0)
ddpg = DDPG("MlpPolicy", env, verbose=0)
```

```
# Train agents
```

```
ppo.learn(total_timesteps=3000)
trpo.learn(total_timesteps=3000)
ddpg.learn(total_timesteps=3000)
```

```
models = {
    "PPO": ppo,
    "TRPO": trpo,
    "DDPG": ddp
}
```

```
average_rewards = []
```

```
# Evaluate each agent
```

```
for name, model in models.items():
```

```
    rewards = []
```

```
    for episode in range(5):
```

```
        state, _ = env.reset()
```

```
        total_reward = 0
```

```
        done = False
```

```
        while not done:
```

```
            action, _ = model.predict(
```

```
                state,
```

```
                deterministic=True
```

```
)

state, reward, terminated, truncated, _ = env.step(action)

done = terminated or truncated
total_reward += reward

rewards.append(total_reward)

average = np.mean(rewards)
average_rewards.append(average)

print(
    name,
    "Average Reward:",
    round(average, 2)
)

# Performance graph
plt.figure(figsize=(8, 5))

plt.bar(
    models.keys(),
    average_rewards
)

plt.xlabel("Algorithm")
plt.ylabel("Average Reward")
plt.title("PPO vs TRPO vs DDPG Performance")

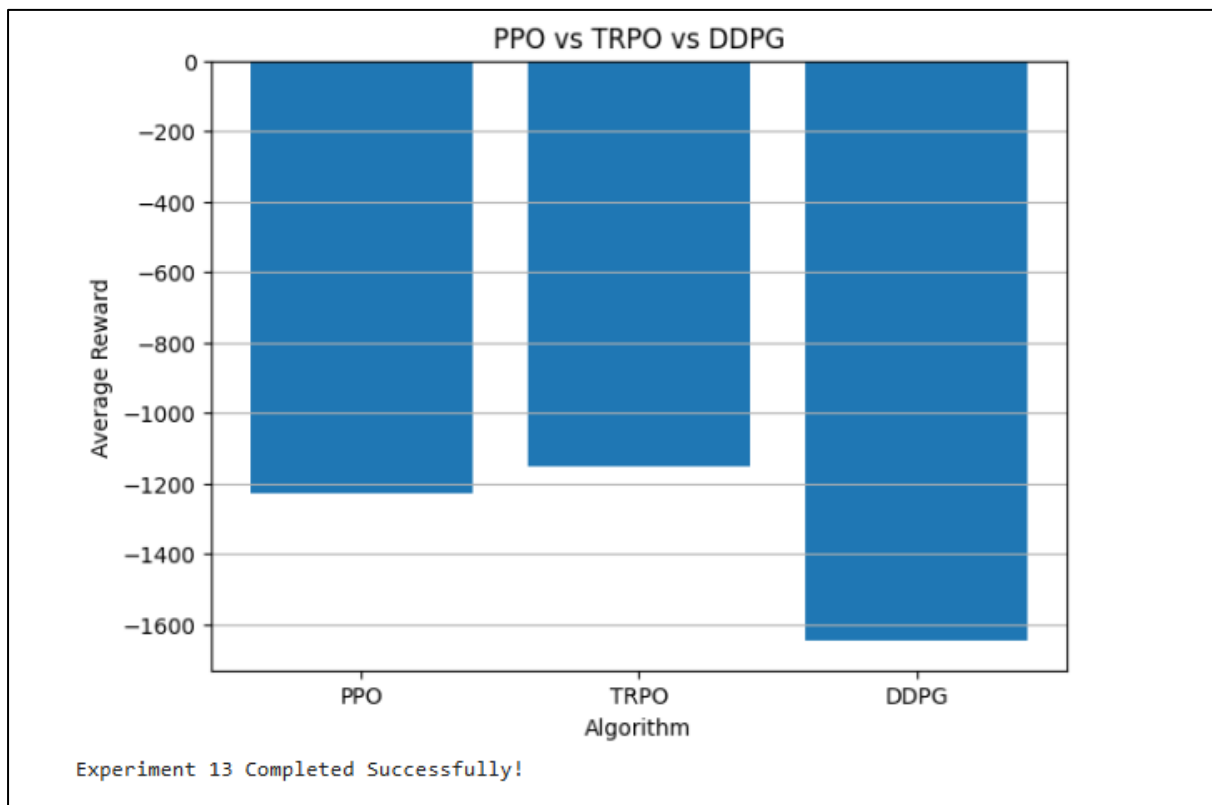
plt.grid(axis="y")
plt.tight_layout()
```

```
plt.show()
```

```
env.close()
```

```
print("\nExperiment 13 Completed Successfully!")
```

Output



Result

PPO, TRPO, and DDPG were successfully implemented and evaluated. Their average rewards were compared using a graphical representation.